Project Design Phase

Problem – Solution Fit Template

Date	4 march 2026
Team ID	SWTID-2026-2725
Project Name	SpendWise
Maximum Marks	2 Marks

Problem – Solution Fit Overview

- The Problem–Solution Fit explains how the project solves a real difficulty faced by beginners when learning backend development. Many new learners find it hard to understand how different backend components such as APIs, authentication, and databases work together in a single project. SpendWise is designed as a simple backend application that demonstrates these concepts in a clear and practical way. It helps learners understand how to build secure APIs and manage data using Node.js, Express, and MongoDB.

Purpose

- The main purpose of the SpendWise project is to provide a beginner-friendly backend application that

helps students learn important backend concepts through practice.

- To simplify the learning process of backend development for beginners.
- To help students understand REST API creation and request handling.
- To demonstrate how authentication works using JWT (JSON Web Token).
- To teach database operations such as creating, updating, and deleting data using MongoDB.
- To provide a structured backend project that can be easily tested using tools like Postman or Thunder Client.

Problem Statement

- Beginners who start learning backend development often face difficulties because learning resources either show only basic API examples or very large real-world applications. Because of this gap, students struggle to understand how authentication systems, database connections, and CRUD operations are implemented together in a real backend project. This lack of practical understanding makes it difficult for learners to build secure and well-structured APIs.

Solution

- SpendWise solves this problem by providing a simple and organized backend project built using Node.js, Express, and MongoDB.

- It includes a user registration and login system secured with JWT authentication.
- Users can manage their data using CRUD operations after logging in.
- Protected routes are implemented using JWT middleware to demonstrate API security.
- The project follows a clean folder structure such as models, routes, controllers, and middleware to show proper backend organization.
- The APIs can be easily tested using tools like Postman or Thunder Client, making the learning process practical and easy for beginners.